C Programming Assignment – Question 10

10. Electricity Bill

Question

Calculate an electricity bill using the assignment's slabs: up to 200 units at 0.50 per unit; up to 400 units at 0.65 per unit plus 100; up to 600 units at 0.80 per unit plus 200; above 600 units at 1.25 per unit plus 425.

Sample Input

```
200
```

Sample Output

```
Rs.100
```

C Program

```
#include <stdio.h>

int main(void) {
    int units;
    double bill;

    printf("Enter units consumed: ");
    scanf("%d", &units);

    if (units < 0) {
        printf("Invalid units\n");
        return 0;
    }

    if (units <= 200) {
        bill = units * 0.50;
    } else if (units <= 400) {
        bill = units * 0.65 + 100;
    } else if (units <= 600) {
        bill = units * 0.80 + 200;
    } else {
        bill = units * 1.25 + 425;
    }

    printf("Rs.%.2f\n", bill);
    return 0;
}
```

Explanation

1. Read the units consumed.

2. Choose the appropriate slab using if/else if conditions.

3. Calculate the bill using that slab's rate and extra charge.

4. Print the bill.

Note

For 200 units, the sample output is Rs.100. The statement uses “paise” for some rates, but this solution interprets the rates as rupees per unit to match the sample.